

SHORT PROJECT OVERVIEW FOR PROSPECTIVE CONSORTIUM PARTNERS

TERRAINS: *Transforming long-tERm field expeRiments into AI-powered soil iNtelligence and decision-support Systems*

Programme: Horizon Europe – Mission Soil

Call: HORIZON-MISS-2026-05-SOIL-04

Type of Action: Innovation Action (IA)

Proposal submission deadline: 23 September 2026

Current stage: The consortium is currently expanding its network and is seeking organisations able to contribute Long-Term Field Experiments (LTEs), long-term datasets, AI expertise, digital technologies or complementary research infrastructures.

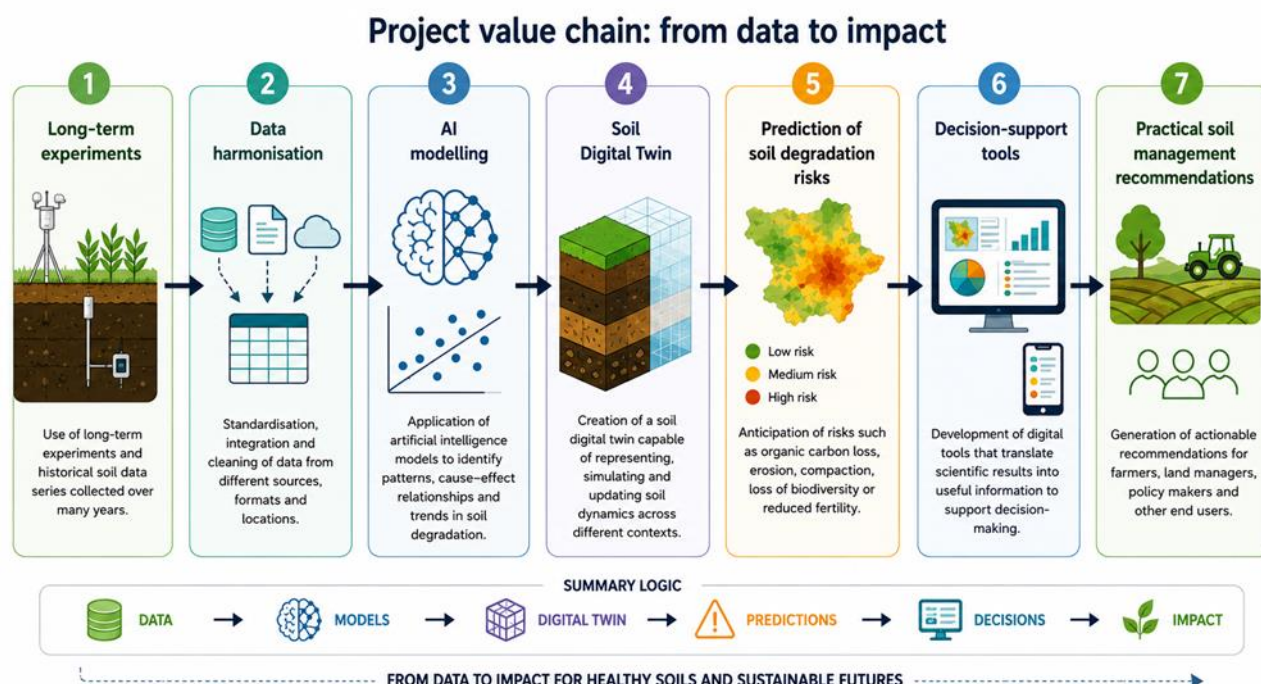
1. Project Vision

TERRAINS aims to develop an interoperable AI-enabled Soil Intelligence framework integrating Long-Term Field Experiments (LTEs), historical soil monitoring datasets, Earth Observation products, climate information and environmental data to support sustainable soil management across diverse European agroecosystems. Building on harmonised and AI-ready datasets, the project will develop a Soil Intelligence Platform, AI models, a scalable Soil Digital Twin and operational Decision Support Systems (DSS) capable of analysing soil–climate–management interactions, predicting soil degradation risks and supporting sustainable land management.

2. Main Objectives

- Harmonise LTEs, historical soil monitoring and environmental datasets across Europe.
- Develop an interoperable Soil Intelligence Platform and AI-ready data infrastructure.
- Build AI models and a scalable Soil Digital Twin for agricultural soils.
- Deliver AI-enabled Decision Support Systems for farmers, advisors and policy makers.
- Validate the developed tools through a network of **>50 Long-Term Field Experiments** representing major European pedo-climatic regions together with selected African datasets.

3. Core Project Components



4. Indicative Work Package Structure

WP	Title
WP1	Project Management
WP2	European & African LTE Network, Data Harmonisation and Integration
WP3	Soil Intelligence Platform & Interoperability
WP4	AI Models & Soil Digital Twin
WP5	AI-enabled Decision Support Systems
WP6	Validation through European & African LTE Network
WP7	Multi-actor Engagement, Living Labs & Capacity Building
WP8	Communication, Exploitation & Sustainability

5. Target Consortium

TERRAINS is building a multidisciplinary consortium that combines expertise in soil science, artificial intelligence, digital technologies and sustainable agricultural management. The project aims to bring together complementary partners capable of supporting the entire innovation chain - from long-term soil observations and data harmonisation to AI development, digital tools and their validation under real farming conditions. Particular priority is given to organisations managing long-term field experiments and historical soil datasets in underrepresented pedo-climatic regions.

The consortium is expected to include:

- Research institutes and experimental stations managing Long-Term Field Experiments (LTEs), national soil monitoring programmes and historical soil datasets across different European pedo-climatic regions.
- Universities and research centres with expertise in artificial intelligence, machine learning, environmental modelling and digital twins for the development of predictive soil intelligence systems.
- Digital innovation hubs and technology providers responsible for developing the Soil Intelligence Platform, ensuring data interoperability and translating scientific results into operational decision-support systems.
- Earth observation and geospatial experts contributing remote sensing products, environmental monitoring data and spatial analysis for large-scale soil assessment.
- Agricultural stakeholders, including farmers' organisations, advisory services and Living Labs, supporting co-development, demonstration and validation of digital solutions under real agricultural conditions.

6. Target Pedo-climatic Coverage

The project aims to establish a representative network of more than 50 LTEs covering major European pedo-climatic regions, contrasting agricultural systems and management practices.

Target regions include:

- Atlantic
- Boreal
- Continental
- Mediterranean
- Alpine
- Pannonian
- Eastern European Chernozems
- African dryland and tropical agroecosystems

Types of Long-Term Experiments (LTEs) Sought

Long-Term Experiments constitute the scientific backbone of TERRAINS. Their long-term observations provide unique evidence on soil responses to management, climate variability and environmental change. Integrating harmonised LTE datasets from diverse pedo-climatic regions will enable robust AI model

development, Digital Twin calibration and validation, and improve the transferability of decision-support tools across Europe and selected African regions.

To establish a representative European LTE network of more than 50 Long-Term Experiments, TERRAINS seeks LTEs covering diverse pedo-climatic regions, agricultural systems and management practices across Europe and selected African regions.

The project is particularly interested in LTEs that include one or more of the following:

- Mineral fertilisation and nutrient management;
- Organic amendments (manure, compost, digestate, biochar);
- Tillage and conservation agriculture;
- Crop rotations and diversification;
- Cover crops and soil conservation practices;
- Organic and conventional farming systems;
- Irrigation and water management;
- Soil carbon and soil organic matter monitoring;
- Climate change adaptation and resilience;
- Soil degradation processes (erosion, salinisation, compaction, acidification);
- Permanent crops (vineyards, orchards), arable systems and grasslands.

Long-term soil monitoring programmes and chronosequences with harmonised historical datasets may also be considered where they provide valuable long-term observations for AI model development and validation. We also welcome completed LTEs, provided that long-term historical datasets remain available for harmonisation and AI applications.

7. Expected Impact

TERRAINS will contribute to the development of a new generation of AI-enabled soil intelligence tools by integrating long-term experimental data, historical soil monitoring, Earth Observation and climate information into a harmonised and interoperable knowledge framework.

The project is expected to deliver:

- A harmonised European and African LTE network, integrating more than 50 Long-Term Field Experiments together with historical soil monitoring datasets covering major pedo-climatic regions.
- An interoperable Soil Intelligence Platform providing AI-ready datasets and supporting data harmonisation of soil datasets across Europe, interoperability and alignment with European initiatives such as EUSO and Mission Soil.
- AI models and a scalable Soil Digital Twin capable of analysing soil–climate–management interactions, predicting soil degradation risks and evaluating alternative management scenarios.
- Operational AI-enabled Decision Support Systems (DSS) to support farmers, advisors, researchers and policy makers with evidence-based recommendations for sustainable soil management.
- Validation of AI models across contrasting pedo-climatic regions, including European Long-Term Experiments and selected African validation sites, demonstrating the robustness and transferability of digital solutions.
- Strengthened European–African scientific collaboration through the integration of complementary datasets, research infrastructures and expertise, contributing to the development of internationally interoperable soil intelligence systems.
- Improved capacity for data-driven soil management, supporting climate resilience, soil health restoration, sustainable agricultural production and evidence-based environmental policies.

Current Consortium Priorities

At this stage, TERRAINS is particularly seeking partners with:

- Long-Term Field Experiments (LTEs)
- Long-term soil monitoring datasets
- Soil data infrastructures
- AI and Machine Learning
- Digital Twins
- Earth Observation
- Digital Innovation Hubs
- African validation sites